

★ COURSE SYLLABUS

Financial Time Series

Advanced Financial Time Series · Yonsei University

For 3rd & 4th-year undergraduates — 16 units, 48 lectures, taught across 14 weeks

From the language of time series (stationarity, ARMA, ARIMA) through volatility (GARCH), the multivariate world (VAR), state space (Kalman, HMM), stochastic processes (SDE), to Bayesian inference (MCMC). Every unit in three parts — theory, deep dive, practice — connecting each idea directly to code and market data.

Class Schedule — When We Meet

Two touchpoints every week

ONLINE · RECORDED

Wednesday

8:00 – 10:00 PM

Recording posted Wednesday evening (8 PM).
Watch anytime before the Friday session.

IN-PERSON · ON CAMPUS

Friday

12:00 – 1:00 PM

Live session for Q&A, discussion, and hands-on practice.

Watch the Wednesday recording first, then bring your questions to Friday's class

What This Course Does

The course in three sentences

1

Translate financial data into the language of models

Recast returns, volatility, and cointegration as stationarity, ARMA, and state-space models — turning raw observation into estimable structure.

2

Treat uncertainty honestly, as probability

Beyond point estimates: GARCH conditional variance, the Kalman update, and Bayesian posteriors quantify how much we do not know.

3

Connect every theory to code and real strategies

Each week's practice applies theory to real market data — pairs trading, VaR, nowcasting, regime switching.

Theory (Part 1) → Deep Dive (Part 2) → Practice (Part 3) — three layers, every week

The Question This Course Answers

**“In financial data that changes over time,
what can we predict — and how large is the uncertainty?”**

Predictability

- What is — and isn't — predictable under market efficiency
- How stationarity and unit roots set the terms of prediction
- Separating conditional mean (ARIMA) from variance (GARCH)

Quantifying Uncertainty

- Kalman filter: update beliefs as each observation arrives
- Bayesian: keep an estimate's full width via the posterior
- Regime switching: capture state changes as probabilities

By the End of This Course

You will have four capabilities

Modeling

Formalize

Cast financial phenomena as ARMA, GARCH, state-space, and SDE models.

e.g. returns $\rightarrow$ GARCH

Inference

Estimate

Estimate parameters and quantify uncertainty via MLE, Kalman, MCMC.

e.g. posterior sampling

Diagnostics

Test

Judge unit roots, cointegration, autocorrelation, model choice by tests.

e.g. ADF · Johansen

Application

Deploy

Backtest pairs, VaR, nowcasting, and regime strategies on real data.

e.g. pairs trading

The Curriculum at a Glance — 4 Modules

16 units build as four blocks, taught in 14 weeks

I. Language of Time Series

Units 1–4

Returns, stationarity, ARMA, ARIMA. The grammar of time series.

Foundations · 12 lectures

II. Volatility & Multivariate

Units 5–7

GARCH, VAR, cointegration. Risk and multi-asset interaction.

Core · 9 lectures

III. State Space & Processes

Units 8–12

Kalman, HMM, SDE, rate models. Latent states and dynamics.

Advanced · 15 lectures

IV. Bayesian

Units 13–16

Bayes, MCMC, state space, model choice + final project.

Integration · 12 lectures

Module I · Language & Stationarity

Units 1–4 · Build the grammar of time series

Unit	Core Topic	Three Parts (Theory · Deep Dive · Practice)
Unit 1	Returns & Markets	Return definitions · Stylized Facts · Market Efficiency
Unit 2	Stationarity & Cointegration	Stationarity · Unit Roots · Cointegration (overview)
Unit 3	ARMA Models	AR · MA · Wold decomposition · ARMA identification · Diagnostics & forecasting
Unit 4	ARIMA & Seasonality	ARIMA · Seasonal SARIMA · ARIMA practice

Module II · Volatility & the Multivariate World

Units 5–7 · Risk and interaction across assets

Unit	Core Topic	Three Parts (Theory · Deep Dive · Practice)
Unit 5	Volatility Models	ARCH · GARCH · VaR and pairs trading
Unit 6	Multivariate VAR	VAR model · Granger causality · Impulse response (IRF) · FEVD
Unit 7	Cointegration & ECM	Cointegration · Cointegration tests · Error-correction (ECM) and pairs

Module III · State Space, Latent States & Processes

Units 8–12 · Track what cannot be seen

Unit	Core Topic	Three Parts (Theory · Deep Dive · Practice)
Unit 8	State Space & Kalman	State-space models · Filtering & smoothing · Midterm synthesis
Unit 9	Kalman Filter Deep Dive	Predict–update · Kalman gain · Pairs and nowcasting
Unit 10	Hidden Markov	Markov hidden states · Forward · Viterbi · Baum-Welch · Regime switching
Unit 11	Stochastic Differential Eqs	Brownian motion · SDE · OU process · OU practice
Unit 12	Short Rates & Term Structure	Vasicek · CIR · Term structure · Discretization & calibration

Module IV · Bayesian Financial Time Series

Units 13–16 · Handle uncertainty through the posterior

Unit	Core Topic	Three Parts (Theory · Deep Dive · Practice)
Unit 13	Bayesian Foundations	Bayes' theorem · Beta-Binomial conjugacy · Intervals & prediction
Unit 14	MCMC	Sample, don't integrate · Metropolis · Gibbs · HMC · NUTS
Unit 15	Bayesian State Space	Why point estimates lie · FFBS · TVP-SV hedge · Bayesian switching
Unit 16	Model Selection & Wrap-up	Three families · Fair comparison · Final project & close

Sixteen Units, Fourteen Teaching Weeks

Weeks 8 and 16 are exams — so two weeks carry two units

Week	Unit · Topic
1	U1 · Returns and Markets
2	U2 · Stationarity and Cointegration
3	U3 + U4 · ARMA and ARIMA
4	U5 · Volatility — ARCH, GARCH, VaR
5	U6 · Multivariate VAR
6	U7 · Cointegration and ECM
7	U8 · State Space and Kalman
8	MIDTERM — units 1–8

Week	Unit · Topic
9	U9 · Kalman Filter Deep Dive
10	U10 · Hidden Markov Models
11	U11 + U12 · SDEs, OU and Short Rates
12	U13 · Bayesian Foundations
13	U14 · MCMC
14	U15 · Bayesian State Space
15	U16 · Model Selection and Close
16	FINAL — units 9–16, capstone due

Weeks 3 and 11 pair two units — ARIMA extends ARMA, and Vasicek is the OU process applied to rates. Units 1–8 sit before the midterm, 9–16 after.

How Each Unit Runs

One unit = three parts

Part 1

Theory

Core concepts and derivations.
Set up the assumptions and intuition.

e.g. GARCH definition

Part 2

Deep Dive

Estimation, testing, extensions.
Add the technical depth.

e.g. estimation · diagnostics

Part 3

Practice

Implement, backtest, and interpret on real market data.

e.g. pairs backtest

A Practice_Applications session runs alongside every part — confirm theory by hand

Assessment & Format

Understanding is proven by hand

Assessment (example)

- Weekly practice assignments — code and interpretation
- Midterm, week 8 — units 1–8, through state space
- Final, week 16 — units 9–16, capstone project due
- Participation — discussion and code review

Operating Principles

- Each part: theory → reproduce in code immediately
- Slides and narrated videos support review
- Formulas are presented with their derivations
- Everything ends in real strategies (pairs · VaR · nowcasting)

What to Prepare

Prerequisites and practice tools

Prerequisites

- Probability & statistics — distributions, expectation, variance, estimation
- Linear algebra — matrix operations, eigenvalues (essential for state space)
- Calculus — differentiation, integration (for SDE and Bayesian work)
- Prior regression experience helps

Practice Tools

- Python — pandas · numpy · statsmodels
- arch · filterpy — GARCH · Kalman filter
- PyMC / NumPyro — Bayesian · MCMC
- Financial data — returns, rates, pair series

Before You Start — FAQ

How to make the most of this course

Q1 Can I keep up if my math is a bit weak?

Difficulty

Q2 Little coding experience — can I do the practice?

Practice

Q3 Which careers and tasks use this directly?

Application

Q4 What is an efficient way to review?

Study

In Summary — How You'll Grow

Where the four modules take you

1

Read time series as the language of models

Diagnose structure and build a forecasting engine with stationarity, ARMA, and ARIMA.

2

Track risk and latent states

Handle volatility, interaction, and unseen states with GARCH, VAR, Kalman, and HMM.

3

Treat uncertainty through the posterior

Preserve the width of estimates with MCMC and Bayesian state space; integrate it all in a final project.

Next lecture — Unit 1, Part 1: starting with Returns